AC9M6N01 • Year 6 Mathematics

Integers on Number Lines and the Cartesian Plane

Represent positive and negative values in financial, temperature, elevation and coordinate contexts

We are learning to…

Students interpret zero as a reference point, order positive and negative integers, use opposites and locate ordered pairs across all four quadrants.

Represent | Reason | Calculate | Interpret

Verify

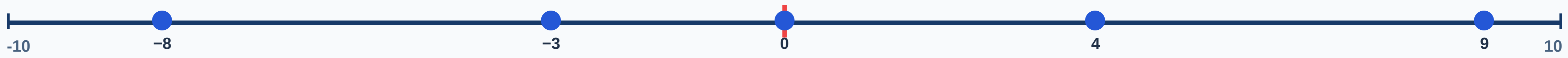
Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

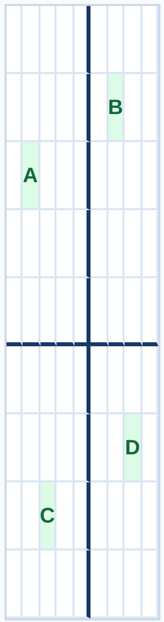
recognise situations, including financial contexts, that use integers; locate and represent integers on a number line and as coordinates on the Cartesian plane

Locate integers relative to zero



Numbers farther right are greater. Negative values are less than zero; their absolute distance from zero is not the same as their signed value.

Connect integer contexts to coordinates



x first, then y

Read an ordered pair as horizontal x-coordinate first, then vertical y-coordinate. The signs identify the quadrant and direction from the origin.

Build and annotate the model

Represent integers on number lines and the cartesian plane and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **−8 is greater than −3 because 8 is greater than 3:** On the number line, −8 lies farther left and is smaller.
- **Negative means a physically impossible quantity:** Integers model debt, below-zero temperature, depth and change.
- **Coordinates are read y then x:** Read x first, then y.
- **Absolute distance confused with signed position:** Distance from zero is non-negative; position keeps its sign.

Quick check

- Order −7, −2, 0 and 5.
- Find the opposite of −9.
- Interpret a bank balance of −$35.
- Plot (−3,2).
- Name the quadrant of (4,−2).

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.